

June 26, 2026

Dear Dr Chevin,

Please find enclosed our revised manuscript, "**Hierarchical structuring of weapon allometry in the European earwig**" (EVO-26-0122.R2), for reconsideration at *Evolution*.

We appreciate the careful and constructive comments provided by you, Associate Editor Dr Emma Sherratt, and Reviewer 1. These comments prompted a substantial restructuring of both the analysis and the interpretation of the study.

Most importantly, we have replaced the previous analyses that pooled the two male morphs with a single unified model in which females, brachylabial males, and macrolabial males are treated as three equivalent phenotypic groups. All condition-specific allometric slopes and all comparisons among groups are now estimated from this model. This approach avoids treating males as a homogeneous category and permits direct comparisons between females and each male morph under a common residual structure.

The revised analysis changes the emphasis of the manuscript. We no longer infer a general difference in allometry between females and males. Instead, the results indicate that females and brachylabial males generally exhibit similar hypoallometric relationships, whereas the clearest divergence occurs in macrolabial males. We have revised the title-related framing, Results, figures, tables, and Discussion to ensure that our conclusions accurately reflect this pattern and do not imply a single, stable male allometric slope.

We have also broadened the manuscript's evolutionary context. The revised Introduction and Discussion now connect weapon allometry to the origin and evolution of sexual dimorphism, the evolution of exaggerated male traits from homologous structures shared with females, and the differentiation of alternative male phenotypes. We have added relevant literature and have softened several causal and evolutionary interpretations where the data support phenotypic patterns rather than direct inferences about historical processes.

A detailed, point-by-point response to all editorial and reviewer comments accompanies the revised manuscript. We have also provided clean and marked versions of the main text, together with the revised supplementary material and associated files.

Thank you for considering our revised manuscript. We hope that the substantial changes have addressed the remaining concerns and strengthened the manuscript's contribution to the study of allometry, sexual dimorphism, and alternative reproductive phenotypes.

Sincerely,



Clint Kelly
on behalf of my co-author